

DATA 260 – HW 3

7- Community Sports League Fixtures

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Hardware: MacBook Pro, Apple M2, 8 GB RAM, macOS

Local model used: qwen2.5:3b (run locally through Ollama)

GitHub repository: <https://github.com/villyalsriprajeet-cmd/data260-3319>

Tagged commit hash: e6a925df2da811cbba91cb37c2c6aa55b12d017d

Section 0 - Personal Configuration and Domain

Value	Definition
SID4	3319
PORT_BASE	8619
PREFIX	s3319
SEED	3319
VERIFY_SEED	263319
DOMAIN_ID	7

Part 1: FastAPI application

I built a FastAPI authentication system on top of my HW2 fixtures app in the same code/api project. It has user login and logout, session management that keeps the user logged in, a protected dashboard that only a logged in user can reach, and Bootstrap styling on every page.

1. Routes - Handled Separately in a router:

All four routes live in their own router file, code/api/routers/auth.py, and main.py loads them with one line, app.include_router(auth_router). The shared pieces (templates, the demo users, the fixtures list and helpers) sit in code/api/store.py so main.py and the router can share them without importing each other.

```
@router.get("/")
def home(request: Request, q: str = ""):
    return templates.TemplateResponse(request, "home.html", {"user": current_user(request), "fixtures": filter_fixtures(q), "q": q})

@router.get("/login")
def login_page(request: Request, error: int = 0):
    if current_user(request): # already logged in, skip the form
        return RedirectResponse(url="/dashboard", status_code=303)
    return templates.TemplateResponse(request, "login.html", {"user": None, "error": bool(error)})

@router.post("/login")
def login(request: Request, username: str = Form(...), password: str = Form(...)):
    if VALID_USERS.get(username) == password:
        request.session["user"] = username # start the session
        return RedirectResponse(url="/dashboard", status_code=303)
    return RedirectResponse(url="/login?error=1", status_code=303) # wrong credentials

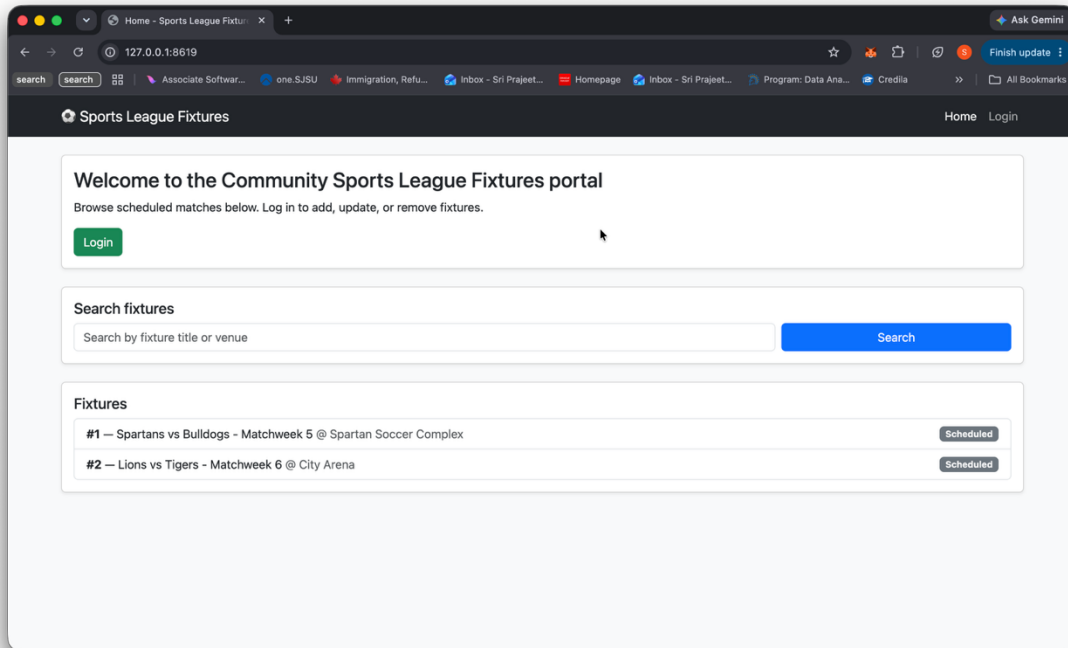
@router.get("/dashboard")
def dashboard(request: Request, q: str = ""):
    if not current_user(request): # protected: must be logged in
        return RedirectResponse(url="/login", status_code=303)
    return templates.TemplateResponse(request, "dashboard.html", {"user": current_user(request), "fixtures": filter_fixtures(q), "q": q})

@router.get("/logout")
def logout(request: Request):
    request.session.clear() # destroy the session
    return RedirectResponse(url="/", status_code=303)
```

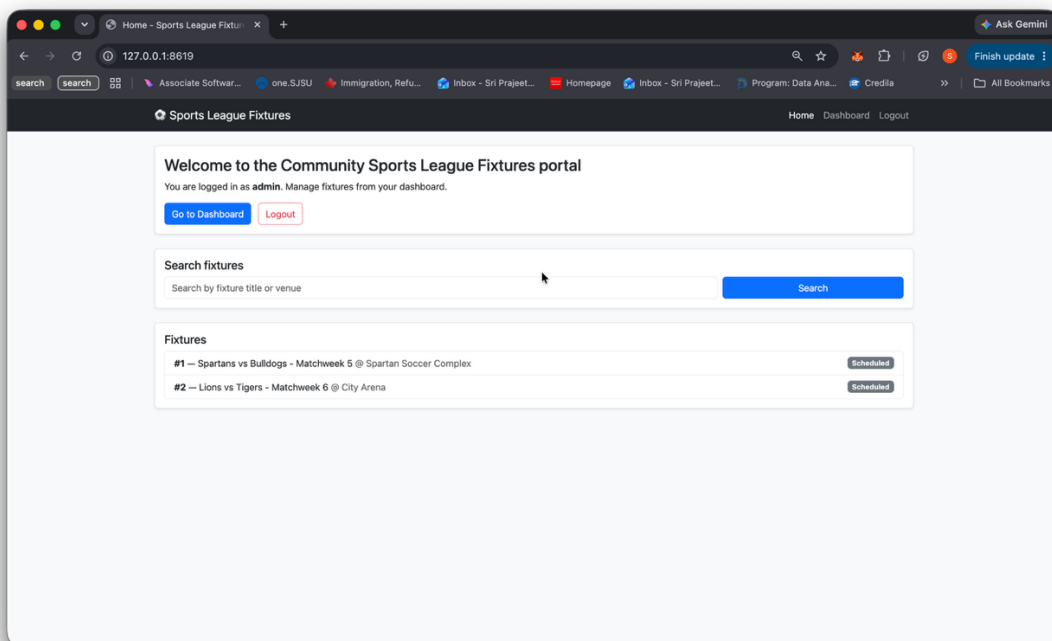
Home Page (/) :

The home page shows a welcome message for the domain. When the user is not logged in it shows a link to the login page and when the user is logged in it shows links to the dashboard and logout.

```
@router.get("/")
def home(request: Request, q: str = ""):
    return templates.TemplateResponse(request, "home.html", {"user": current_user(request), "fixtures": filter_fixtures(q), "q": q})
```



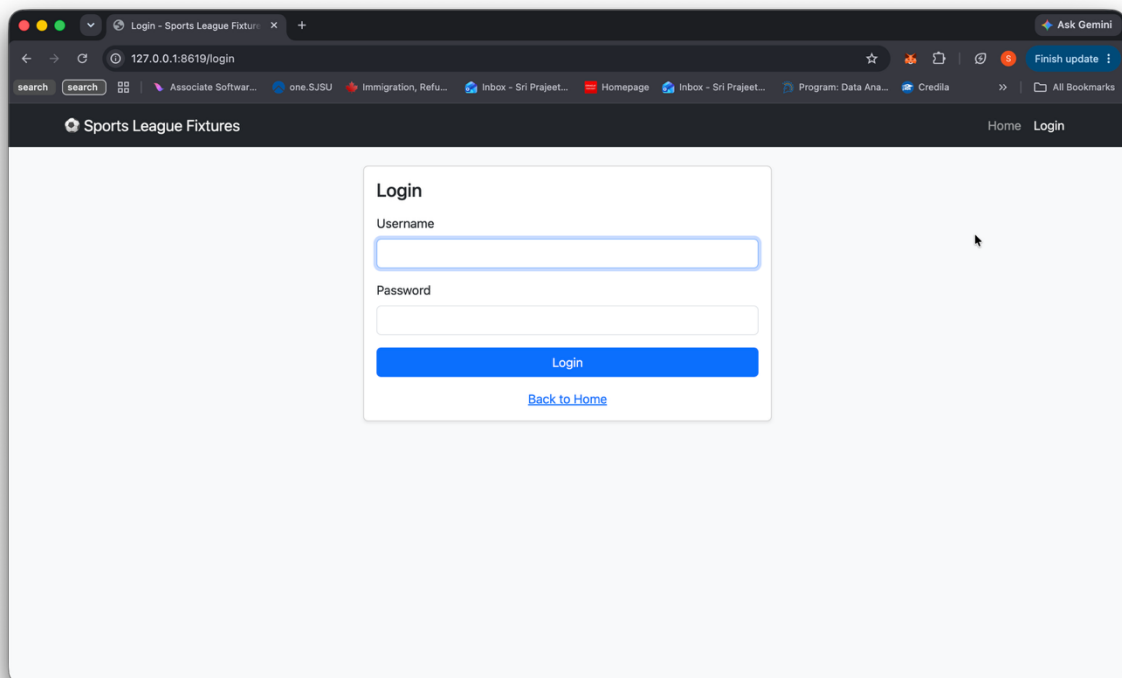
And when the user is logged in the navbar shows dashboard and logout.

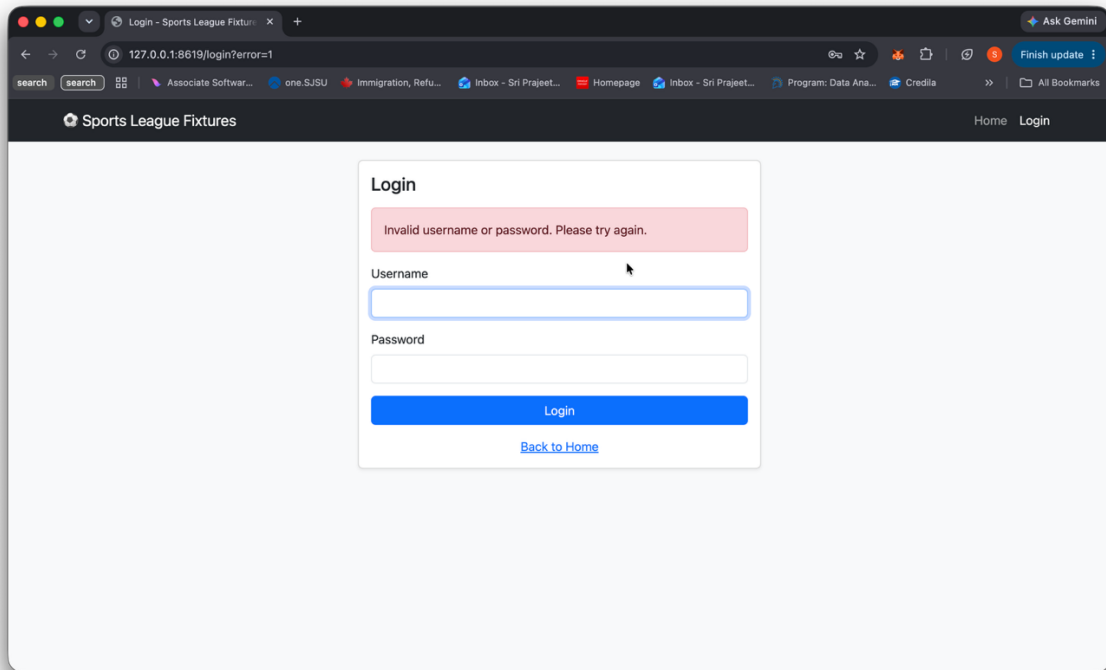


Login Page (/login):

GET /login shows the form, the POST /login validates the credentials, logs the user in and redirects to the dashboard on success, or redirects back with error=1 on failure.

```
@router.get("/login")
def login_page(request: Request, error: int = 0):
    if current_user(request): # already logged in, skip the form
        return RedirectResponse(url="/dashboard", status_code=303)
    return templates.TemplateResponse(request, "login.html", {"user": None, "error": bool(error)})
@router.post("/login")
def login(request: Request, username: str = Form(...), password: str = Form(...)):
    if VALID_USERS.get(username) == password:
        request.session["user"] = username # start the session
        return RedirectResponse(url="/dashboard", status_code=303)
    return RedirectResponse(url="/login?error=1", status_code=303) # wrong credentials
```

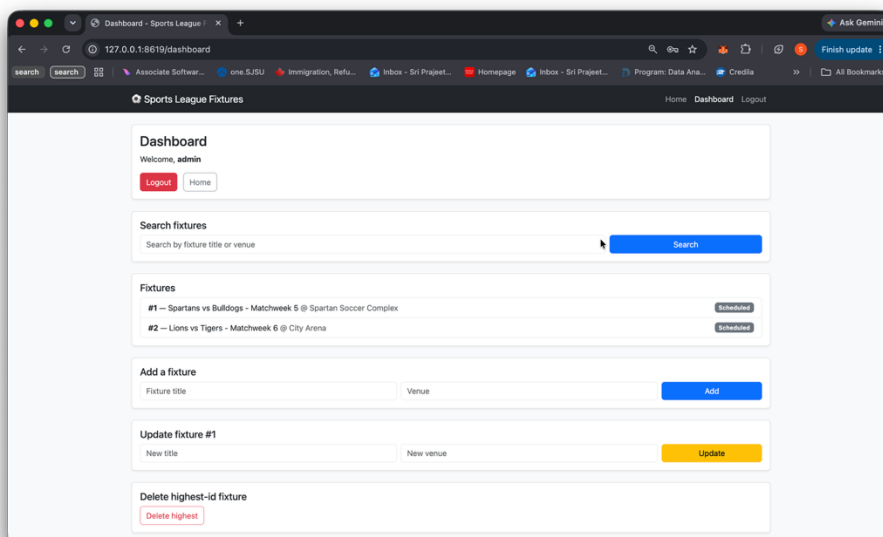




Dashboard Page (/dashboard):

Shows a welcome message with the users name and a logout link. It is protected, the first line checks the session and redirects to /login if no one is logged in.

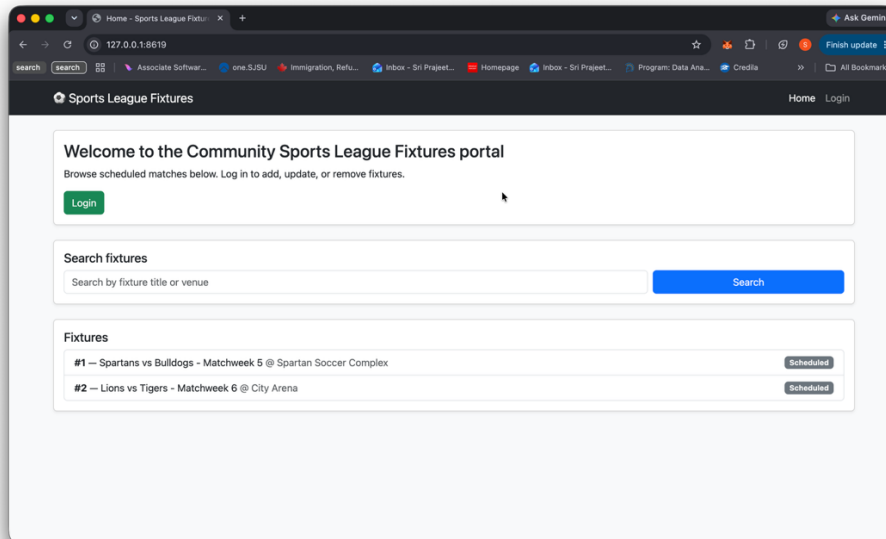
```
@router.get("/dashboard")
def dashboard(request: Request, q: str = ""):
    if not current_user(request): # protected: must be logged in
        return RedirectResponse(url="/login", status_code=303)
    return templates.TemplateResponse(request, "dashboard.html", {"user": current_user(request), "fixtures": filter_fixtures(q), "q": q})
```



Logout (/logout):

Destroys the session and redirects the user back to the home page.

```
@router.get("/logout")
def logout(request: Request):
    request.session.clear() # destroy the session
    return RedirectResponse(url="/", status_code=303)
```

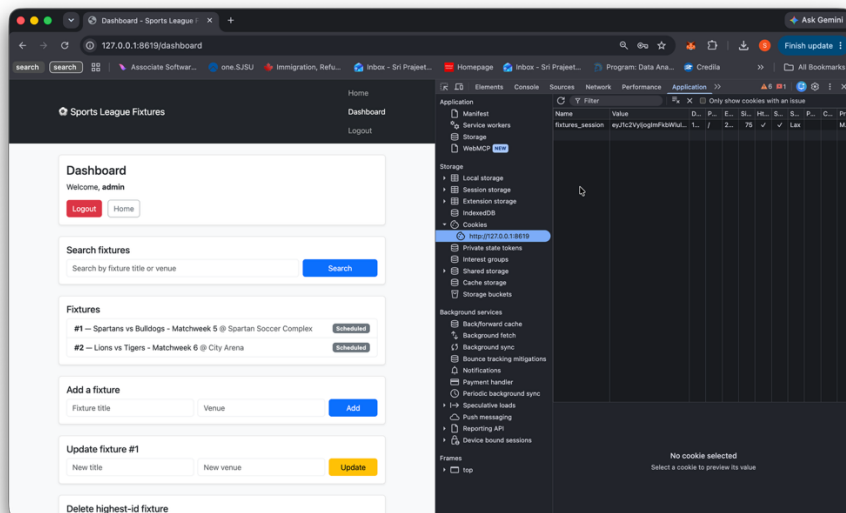
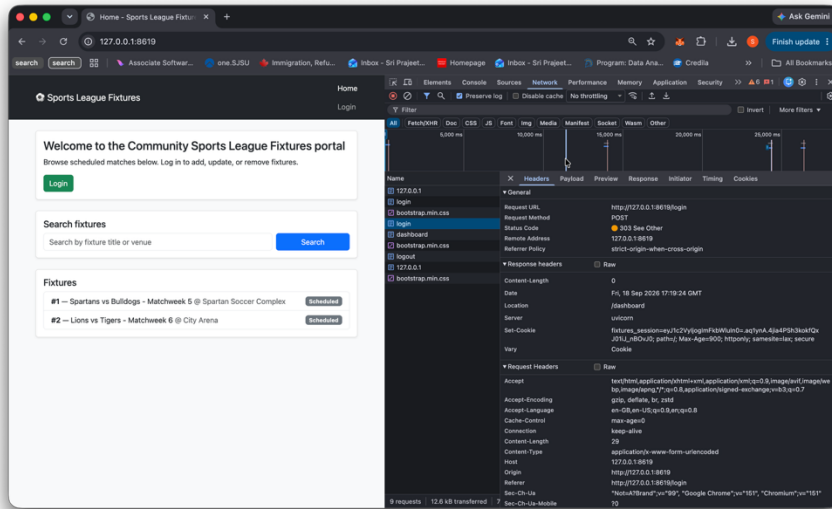


2. Session Management

Sessions use Starlette's SessionMiddleware, configured in code/api/main.py. On login the username is stored in the session, and the cookie is signed and given three attributes.

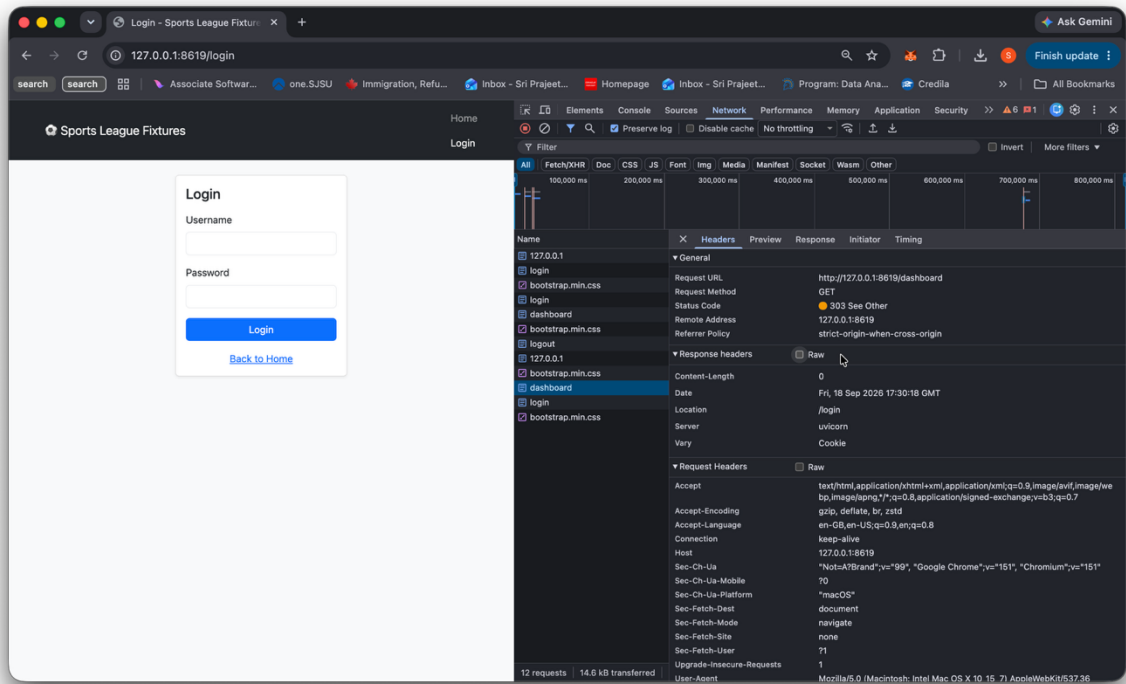
```
app.add_middleware(
    SessionMiddleware,
    secret_key=SECRET_KEY,
    session_cookie="fixtures_session",
    max_age=SESSION_MAX_AGE,
    same_site="lax",
    https_only=COOKIE_SECURE,
)
```

Secure cookie



Idle session timeout

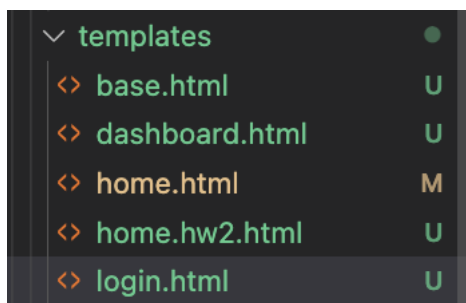
```
PORT_BASE = 8619
SECRET_KEY = os.getenv("SECRET_KEY", "hw3-dev-only-secret-s3319")
SESSION_MAX_AGE = int(os.getenv("SESSION_MAX_AGE", "900")) # idle timeout in seconds
COOKIE_SECURE = os.getenv("COOKIE_SECURE", "1") not in ("0", "false", "False") # Secure attribute
app = FastAPI(title="Community Sports League Fixtures - HW3")
```



3. Styling with Bootstrap

Every page is styled with Bootstrap and is responsive. `base.html` carries the Navbar and is extended by the other pages. The app uses the Bootstrap components as required in the assignment.

Templates Directory



Auth.py

```
code > api > routers > auth.py > logout
1 # HW3 Part 1 auth routes (home, login, dashboard, logout)
2 from fastapi import APIRouter, Request, Form
3 from fastapi.responses import RedirectResponse
4 from ..store import templates, current_user, filter_fixtures, VALID_USERS
5 router = APIRouter()
6 @router.get("/")
7 def home(request: Request, q: str = ""):
8     return templates.TemplateResponse(request, "home.html", {"user": current_user(request), "fixtures": filter_fixtures(q), "q": q})
9 @router.get("/login")
10 def login_page(request: Request, error: int = 0):
11     if current_user(request): # already logged in, skip the form
12         return RedirectResponse(url="/dashboard", status_code=303)
13     return templates.TemplateResponse(request, "login.html", {"user": None, "error": bool(error)})
14 @router.post("/login")
15 def login(request: Request, username: str = Form(...), password: str = Form(...)):
16     if VALID_USERS.get(username) == password:
17         request.session["user"] = username # star (parameter) status_code: int
18         return RedirectResponse(url="/dashboard", status_code=303)
19     return RedirectResponse(url="/login?error=1", status_code=303) # wrong credentials
20 @router.get("/dashboard")
21 def dashboard(request: Request, q: str = ""):
22     if not current_user(request): # protected: must be logged in
23         return RedirectResponse(url="/login", status_code=303)
24     return templates.TemplateResponse(request, "dashboard.html", {"user": current_user(request), "fixtures": filter_fixtures(q), "q": q})
25 @router.get("/logout")
26 def logout(request: Request):
27     request.session.clear() # destroy the session
28     return RedirectResponse(url="/", status_code=303)
```

Part 2: Compare Three LlamaIndex Chunking Techniques (Retrieval-Only RAG)

I implemented three chunking techniques in LlamaIndex, built an in memory vector index for each, and compared how well they retrieve. For a shared set of questions I print the query embedding and the retrieval output for each technique, then argue which one is best for my corpus.

Techniques implemented

1. Token-based chunking - TokenTextSplitter
2. Semantic chunking - SemanticSplitterNodeParser
3. Sentence-window chunking – SentenceWindowNodeParser

Dataset

As a warm up I ran the pipeline on the Tiny Shakespeare raw text first, just to confirm all three chunkers, the index, and retrieval worked before the graded run.

```
TECHNIQUE: token
=====
chunks=1424 avg_chunk_len=897.7 chars

[w1] To be or not to be?
query embedding dim=384 first8=[0.0357, 0.047, -0.0157, 0.0448, 0.0322, 0.0291, 0.1086, -0.1071]
rank store cosine len source preview
1 0.2675 0.3138 928 tinyshakespeare.txt you have been a rod to her friends; you have not indeed
2 0.25 0.2223 892 tinyshakespeare.txt do this feat, Achieve the elder, set the younger free F
3 0.2435 0.2865 983 tinyshakespeare.txt wisdom, Cannot conclude but by the yea and no Of genera
shapes: query=(1, 384) docs=(3, 384) latency=105.67 ms
```

```

TECHNIQUE: semantic
=====
chunks=624 avg_chunk_len=1787.5 chars

[w1] To be or not to be?
query embedding dim=384 first8=[0.0357, 0.047, -0.0157, 0.0448, 0.0322, 0.0291, 0.1086, -0.1071]
rank store cosine len source preview
1 0.2756 0.2957 232 tinyshakespeare.txt my life is my foe's debt. BENVOLIO: Away, begone; the s
2 0.2507 0.2128 650 tinyshakespeare.txt No; I would tell what 'twere to be a judge, And what a
3 0.2454 0.2427 402 tinyshakespeare.txt trouble us not. GONZALO: Good, yet remember whom thou h
shapes: query=(1, 384) docs=(3, 384) latency=62.11 ms

TECHNIQUE: sentence_window
=====
chunks=12453 avg_chunk_len=89.6 chars

[w1] To be or not to be?
query embedding dim=384 first8=[0.0357, 0.047, -0.0157, 0.0448, 0.0322, 0.0291, 0.1086, -0.1071]
rank store cosine len source preview
1 0.3563 0.4033 83 tinyshakespeare.txt LEONTES: At the queen's be't: 'good' should be pertinen
2 0.3165 0.3641 283 tinyshakespeare.txt Officer: HERMIONE: Since what I am to say must be but t
3 0.3106 0.3389 146 tinyshakespeare.txt LORD ROSS: Be confident to speak, Northumberland: We th
shapes: query=(1, 384) docs=(3, 384) latency=644.49 ms

Wrote 9 rows to reports/hw03/raw/ (retrieval_rows.jsonl, .csv) + technique_stats.json
Run finished: 2026-09-19T09:26:34.617254

```

My actual dataset is 303 KB of local snapshots of public Wikipedia season/fixtures articles for my domain (community sports league fixtures). build_corpus.py downloads the rendered pages, converts the tables to text (so the fixtures and results are kept), and writes SOURCES.md and CORPUS_MANIFEST.json with each file's byte size and SHA-256 hash.

The fetch and manifest

```

def fetch_text(title): # rendered HTML via the parse API, then converted to text
    params = {"action": "parse", "page": title, "prop": "text",
              "redirects": "1", "format": "json", "formatversion": "2"}
    url = WIKI_API + "?" + urllib.parse.urlencode(params)
    req = urllib.request.Request(url, headers={"User-Agent": UA})
    with urllib.request.urlopen(req, timeout=90) as r:
        html = json.load(r)["parse"]["text"]
    return html_to_text(html)

```

```

manifest.append({"title": title, "local_file": rel, "url": wiki_url(title),
                "access_date": today, "bytes": size, "sha256": sha256(fpath)})

```

```

(data260) villyalsriprajeet@Mac data260-3319 % python code/rag/build_corpus.py
Fetching: 2023-24 Premier League ...
  saved data/corpus/2023_24_premier_league.txt (73431 bytes)
Fetching: 2024 Major League Soccer season ...
  saved data/corpus/2024_major_league_soccer_season.txt (91660 bytes)
Fetching: 2023-24 La Liga ...
  saved data/corpus/2023_24_la_liga.txt (45540 bytes)
Fetching: 2023-24 EFL Championship ...
  saved data/corpus/2023_24_efl_championship.txt (47729 bytes)
Fetching: 2024-25 UEFA Champions League ...
  saved data/corpus/2024_25_uefa_champions_league.txt (52245 bytes)

Total corpus size: 310605 bytes (303.3 KB)
OK: corpus >= 200 KB.
Warm-up saved: data/tinyshakespeare.txt (1115394 bytes)

```

Environment and Setup

I installed llama index, llama index embeddings huggingface, sentence transformers, faiss-cpu, numpy and pandas, and used the public sentence transformers/all-MiniLM-L6-v2 model (384-dim) pulled from HuggingFace. I pinned numpy 1.26.4 and transformers 4.41.2 to match torch 2.2.2 so the model would load.

One retrieval only pipeline per technique

1. Chunking

Each technique is a different LlamaIndex node parser. Token uses a token chunk_size and overlap; semantic uses a buffer_size and the embed model to find coherent boundaries. The sentence window splits into single sentences and keeps neighbour sentences in metadata.

```
def build_parsers(embed):
    return {
        "token": TokenTextSplitter(chunk_size=256, chunk_overlap=32),
        "semantic": SemanticSplitterNodeParser(buffer_size=1, breakpoint_percentile_threshold=95, embed_model=embed),
        "sentence_window": SentenceWindowNodeParser.from_defaults(window_size=3),
    }
```

2. Indexing

For each set of chunks I build a VectorStoreIndex, which uses LlamaIndex in memory SimpleVectorStore by default, so everything stays local and fast.

```
index = VectorStoreIndex(nodes) # in memory SimpleVectorStore
retriever = index.as_retriever(similarity_top_k=k)
```

3. Retrieval-only function

The helper embeds the query, retrieves the top k chunks, and for each one prints the store score, an explicit cosine similarity, the chunk length and a short preview, plus the shapes of the query and stacked document vectors.

```
qv = embed.get_query_embedding(question)
t0 = time.perf_counter()
hits = retriever.retrieve(question)
latency_ms = round((time.perf_counter() - t0) * 1000, 2)
print(f"\n[{qid}] {question}")
print(f" query embedding dim={len(qv)} first8={round(x,4) for x in qv[:8]}")
print(f" {'rank':<4} {'store':<9} {'cosine':<9} {'len':<6} {'source':<30} preview")
doc_vecs = []
for rank, h in enumerate(hits, 1):
    txt = h.node.get_content()
    dv = embed.get_text_embedding(txt); doc_vecs.append(dv)
    cs = round(cosine(qv, dv), 4)
    src = h.node.metadata.get("source", "")
    preview = re.sub(r"\s+", " ", txt)[:160]
    rows.append({"technique": name, "qid": qid, "question": question, "rank": rank,
                "store_score": round(float(h.score or 0.0), 4), "cosine_sim": cs,
                "chunk_len": len(txt), "source": src,
                "in_expected_source": (src == expected), "latency_ms": latency_ms,
                "preview": preview})
print(f" {'rank':<4} {round(float(h.score or 0.0),4):<9} {cs:<9} {len(txt):<6} {os.path.basename(src):<30} {preview[:55]}")
qvs, dvs = np.array([qv]), np.array(doc_vecs)
print(f" shapes: query={qvs.shape} docs={dvs.shape} latency={latency_ms} ms")
```

Token output

```
• (data260) villyalsriprajee@Mac data260-3319 % python code/rag/rag_compare.py 2>61 | tee reports/hw03/RUN_LOG.txt
/Users/villyalsriprajee/miniconda3_x86/envs/data260/lib/python3.12/site-packages/huggingface_hub/file_download.py:1132: FutureWarning: 'resume_download' is deprecated and will be removed in version 1.0.0. Downloads always resume when possible. If you want to force a new download, use 'force_download=True'.
  warnings.warn(
Run started: 2026-09-18T21:00:32.149274

=====
TECHNIQUE: token
=====
chunks=499 avg_chunk_len=700.3 chars

[q1] Who were the champions of the 2023-24 Premier League?
query embedding dim=384 first8=[-0.0356, -0.0307, -0.0734, -0.015, 0.0048, 0.1213, -0.0162, 0.0243]
rank store cosine len source preview
1 0.6328 0.6258 1013 2023_24_premier_league.txt were the first team to have been promoted from non-League
2 0.6153 0.6079 956 2023_24_premier_league.txt "When does the 2023/24 Premier League season start? Key
3 0.6128 0.6244 865 2023_24_premier_league.txt Seasons Awards Hall of Fame | Finances | Richest clubs
4 0.6088 0.5796 866 2023_24_premier_league.txt Football season in England For other uses, see 2023-24
5 0.5857 0.6208 697 2023_24_premier_league.txt September 2023 | Erling Haaland | Manchester City | Ful
shapes: query=(1, 384) docs=(5, 384) latency=48.48 ms

[q2] Which club won the 2023-24 La Liga title?
query embedding dim=384 first8=[-0.023, -0.0303, -0.0555, -0.0101, -0.0085, 0.0728, -0.0428, 0.0679]
rank store cosine len source preview
1 0.7149 0.7124 832 2023_24_la_liga.txt 2022-23 2024-25 - The 2023-24 La Liga, also known as La
2 0.6539 0.6682 769 2023_24_la_liga.txt "OFFICIAL: LaLiga defines the start and end dates for t
3 0.64 0.6273 795 2023_24_la_liga.txt Retrieved 2024-06-25. ^ Menzies, John (2023-05-28). "Gr
4 0.638 0.6117 912 2023_24_la_liga.txt 1) Points; 2) Head-to-head points; 3) Head-to-head goal
5 0.6255 0.6421 689 2023_24_la_liga.txt y EA SPORTS desvelan los 15 mejores jugadores de la tem
shapes: query=(1, 384) docs=(5, 384) latency=37.35 ms

[q3] Which team won the 2023-24 EFL Championship and were promoted as champions?
query embedding dim=384 first8=[-0.0538, 0.0001, -0.0365, -0.0246, 0.0221, 0.0875, -0.0088, 0.0241]
rank store cosine len source preview
1 0.594 0.2053 349 2023_24_efl_championship.txt 1914-15 1915-16 1916-17 1917-18 1918-19 1919-20 1920-21
2 0.5842 0.5263 271 2023_24_efl_championship.txt 1982-83 1983-84 1984-85 1985-86 1986-87 1987-88 1988-89
3 0.5675 0.5366 869 2023_24_efl_championship.txt (23 September 2023) Southampton 5-3 Huddersfield Town (
4 0.5651 0.2919 349 2023_24_efl_championship.txt 1948-49 1949-50 1950-51 1951-52 1952-53 1953-54 1954-55
5 0.5635 0.5979 918 2023_24_efl_championship.txt the EFL Championship under its current title and the 32
shapes: query=(1, 384) docs=(5, 384) latency=36.37 ms

[q4] At which stadium was the 2024-25 UEFA Champions League final played?
query embedding dim=384 first8=[0.0327, 0.0862, 0.0015, -0.0414, 0.0141, 0.0107, -0.0628, 0.0253]
rank store cosine len source preview
1 0.7091 0.7115 646 2024_25_uefa_champions_league.txt and 30 April, and the second legs were played on 6 and
2 0.6936 0.6561 1050 2024_25_uefa_champions_league.txt Doué (Paris Saint-Germain) | - 2023-24 2025-26 + Intern
3 0.6854 0.698 215 2024_25_uefa_champions_league.txt Commons has media related to 2024-25 UEFA Champions Lea
4 0.6575 0.6386 889 2024_25_uefa_champions_league.txt UEFA. Union of European Football Associations. Retrieve
5 0.6118 0.6315 891 2024_25_uefa_champions_league.txt UEFA Women's Champions League 2024-25 UEFA Youth League
shapes: query=(1, 384) docs=(5, 384) latency=37.12 ms

[q5] Which club won the 2024 MLS Supporters' Shield?
query embedding dim=384 first8=[0.0103, -0.007, -0.0537, 0.0306, 0.0789, 0.0402, -0.0008, 0.0401]
rank store cosine len source preview
1 0.6828 0.6736 1067 2024_major_league_soccer_season.txt Eastern and Western conferences. This was the first sea
2 0.678 0.6424 948 2024_major_league_soccer_season.txt U.S. Open Cup winners; (V) Canadian Championship winner
3 0.6375 0.5992 727 2024_major_league_soccer_season.txt Chicago Fire FC | 34 | 7 | 18 | 9 | 40 | 62 | -22 | 30
4 0.616 0.5777 817 2024_major_league_soccer_season.txt | 7 | 51 | 66 | -15 | 31 | 14 | San Jose Earthquakes |
5 0.6087 0.5645 657 2024_major_league_soccer_season.txt spot in Leagues Cup was given to San Diego FC. Overall
shapes: query=(1, 384) docs=(5, 384) latency=40.64 ms
```

Semantic output

```
TECHNIQUE: semantic
=====
chunks=167 avg_chunk_len=1821.3 chars

[q1] Who were the champions of the 2023-24 Premier League?
query embedding dim=384 first8=[-0.0356, -0.0307, -0.0734, -0.015, 0.0048, 0.1213, -0.0162, 0.0243]
rank store cosine len source preview
1 0.5941 0.5837 996 2023_24_premier_league.txt Retrieved 14 May 2024. ^ "Dates for 2023/24 season conf
2 0.5639 0.517 20465 2023_24_premier_league.txt Football season in England For other uses, see 2023-24
3 0.5382 0.4834 1671 2023_24_premier_league.txt Archived from the original on 21 February 2020. Retriev
4 0.5344 0.525 1169 2023_24_premier_league.txt Sport UK. London. Archived from the original on 21 May
5 0.5331 0.5109 25735 2023_24_efl_championship.txt Football league season EFL Championship | Season | 2023
shapes: query=(1, 384) docs=(5, 384) latency=65.55 ms

[q2] Which club won the 2023-24 La Liga title?
query embedding dim=384 first8=[-0.023, -0.0303, -0.0555, -0.0101, -0.0085, 0.0728, -0.0428, 0.0679]
rank store cosine len source preview
1 0.6716 0.6475 384 2023_24_la_liga.txt 2 August 2022. Retrieved 27 April 2023. ^ "EA SPORTS FC
2 0.644 0.6107 482 2023_24_la_liga.txt Retrieved 4 July 2018. ^ "LaLiga bate record histórico
3 0.5952 0.6046 7263 2023_24_la_liga.txt 93rd season of La Liga Football league season La Liga |
4 0.5939 0.5442 292 2023_24_la_liga.txt Retrieved 1 July 2019. ^ "Facilities". ^ "Montilivi se
5 0.5796 0.5915 390 2023_24_la_liga.txt Retrieved 29 May 2022. ^ "The Stadium". Athletic Bilbao
shapes: query=(1, 384) docs=(5, 384) latency=79.95 ms

[q3] Which team won the 2023-24 EFL Championship and were promoted as champions?
query embedding dim=384 first8=[-0.0538, 0.0001, -0.0365, -0.0246, 0.0221, 0.0875, -0.0088, 0.0241]
rank store cosine len source preview
1 0.561 0.1708 43 2023_24_efl_championship.txt Archived from the original on 1 July 2021.
2 0.5035 0.2075 15 2023_24_efl_championship.txt 17 April 2024.
3 0.4947 0.4289 413 2023_24_efl_championship.txt Retrieved 8 April 2022. ^ "Hull City 4-2 Sheffield Wedn
4 0.4899 0.4935 384 2023_24_la_liga.txt 2 August 2022. Retrieved 27 April 2023. ^ "EA SPORTS FC
5 0.4748 0.4668 8418 2023_24_efl_championship.txt 9 April 2024. Retrieved 9 April 2024. ^ Aloia, Andrew (
shapes: query=(1, 384) docs=(5, 384) latency=31.05 ms

[q4] At which stadium was the 2024-25 UEFA Champions League final played?
query embedding dim=384 first8=[0.0327, 0.0862, 0.0015, -0.0414, 0.0141, 0.0107, -0.0628, 0.0253]
rank store cosine len source preview
1 0.6709 0.6606 9591 2024_25_uefa_champions_league.txt European football tournament 2024-25 UEFA Champions Lea
2 0.6025 0.6001 162 2024_25_uefa_champions_league.txt Union of European Football Associations. 31 May 2025. R
3 0.5983 0.5753 368 2024_25_uefa_champions_league.txt 10 July 2023. Archived from the original on 16 March 20
4 0.5818 0.5806 785 2024_25_uefa_champions_league.txt Union of European Football Associations. 1 June 2025. R
5 0.5752 0.5499 241 2024_25_uefa_champions_league.txt Retrieved 27 August 2024. ^ "UEFA Club Competitions 202
shapes: query=(1, 384) docs=(5, 384) latency=26.02 ms

[q5] Which club won the 2024 MLS Supporters' Shield?
query embedding dim=384 first8=[0.0103, -0.007, -0.0537, 0.0306, 0.0789, 0.0402, -0.0008, 0.0401]
rank store cosine len source preview
1 0.552 0.5224 169 2024_major_league_soccer_season.txt ^ McCaskill, Steve (November 17, 2022). "MLS clubs to p
2 0.5436 0.5335 1742 2023_24_efl_championship.txt ESPN. Retrieved 4 August 2023. ^ "Undefeated named Blue
3 0.5314 0.4883 165 2024_major_league_soccer_season.txt Retrieved June 24, 2024. ^ Boehm, Charles (July 1, 2024
4 0.5299 0.506 384 2023_24_la_liga.txt 2 August 2022. Retrieved 27 April 2023. ^ "EA SPORTS FC
5 0.522 0.475 145 2024_major_league_soccer_season.txt Archived from the original on February 21, 2024. Retriev
shapes: query=(1, 384) docs=(5, 384) latency=36.49 ms
```

Sentence window output

```
TECHNIQUE: sentence_window
=====
chunks=3202 avg_chunk_len=95.0 chars

[q1] Who were the champions of the 2023-24 Premier League?
query embedding dim=384 first8=[-0.0356, -0.0307, -0.0734, -0.015, 0.0048, 0.1213, -0.0162, 0.0243]
rank store cosine len source preview
1 0.6417 0.5054 153 2023_24_premier_league.txt Teams [edit] Twenty teams competed in the league – the
2 0.6102 0.1392 54 2023_24_premier_league.txt Archived (PDF) from the original on 1 September 2023.
3 0.6098 0.554 175 2023_24_premier_league.txt They replaced Leicester City, Leeds United and Southamp
4 0.6095 0.6461 49 2023_24_premier_league.txt ^ Premier League Handbook: Season 2023/24 (PDF).
5 0.6059 0.1585 43 2023_24_premier_league.txt Archived from the original on 21 May 2021.
shapes: query=(1, 384) docs=(5, 384) latency=188.15 ms

[q2] Which club won the 2023-24 La Liga title?
query embedding dim=384 first8=[-0.023, -0.0303, -0.0555, -0.0101, -0.0085, 0.0728, -0.0428, 0.0679]
rank store cosine len source preview
1 0.6603 0.5824 90 2023_24_la_liga.txt Barcelona were the defending champions, having won thei
2 0.6476 0.5652 151 2023_24_la_liga.txt The third and final team relegated to Segunda was Valla
3 0.6448 0.5346 104 2023_24_la_liga.txt This included the two top teams from the Segunda Divisi
4 0.6225 0.5986 178 2023_24_la_liga.txt On 4 May 2024, Real Madrid were officially confirmed as
5 0.618 0.6193 155 2023_24_la_liga.txt ^ https://www.european-football-statistics.co.uk/attn/a
shapes: query=(1, 384) docs=(5, 384) latency=141.71 ms

[q3] Which team won the 2023-24 EFL Championship and were promoted as champions?
query embedding dim=384 first8=[-0.0538, 0.0001, -0.0365, -0.0246, 0.0221, 0.0875, -0.0088, 0.0241]
rank store cosine len source preview
1 0.586 0.5347 54 2023_24_efl_championship.txt ^ "See who made the EFL Team of the Season line-ups".
2 0.5773 0.4376 157 2023_24_efl_championship.txt Team changes [edit] The following teams have changed di
3 0.5663 0.1978 43 2023_24_efl_championship.txt Archived from the original on 3 July 2022.
4 0.5658 0.1889 45 2023_24_efl_championship.txt Archived from the original on 10 July 2023.
5 0.5613 0.1899 44 2023_24_efl_championship.txt Archived from the original on 25 June 2021.
shapes: query=(1, 384) docs=(5, 384) latency=128.8 ms

[q4] At which stadium was the 2024-25 UEFA Champions League final played?
query embedding dim=384 first8=[0.0327, 0.0862, 0.0015, -0.0414, 0.0141, 0.0107, -0.0628, 0.0253]
rank store cosine len source preview
1 0.7538 0.7705 130 2024_25_uefa_champions_league.txt Final [edit] Main article: 2025 UEFA Champions League f
2 0.7089 0.6446 94 2024_25_uefa_champions_league.txt External links [edit] Wikimedia Commons has media relat
3 0.6709 0.6751 1036 2024_25_uefa_champions_league.txt European football tournament 2024-25 UEFA Champions Lea
4 0.6684 0.6693 210 2024_25_uefa_champions_league.txt The winners of the 2023-24 UEFA Champions League and 20
5 0.6444 0.6285 347 2024_25_uefa_champions_league.txt As winners of the Champions League, Paris Saint-Germain
shapes: query=(1, 384) docs=(5, 384) latency=123.93 ms

[q5] Which club won the 2024 MLS Supporters' Shield?
query embedding dim=384 first8=[0.0103, -0.007, -0.0537, 0.0306, 0.0789, 0.0402, -0.0008, 0.0401]
rank store cosine len source preview
1 0.7098 0.5766 82 2024_major_league_soccer_season.txt Overall table [edit] The leading team in this table won
2 0.7096 0.6748 471 2024_major_league_soccer_season.txt Open Cup winners; (V) Canadian Championship winners Not
3 0.6695 0.6307 113 2024_major_league_soccer_season.txt The Columbus Crew were the reigning MLS Cup champions a
4 0.6277 0.6449 51 2024_major_league_soccer_season.txt ^ "FC Cincinnati clinches MLS Supporters' Shield".
5 0.5951 0.5165 134 2024_major_league_soccer_season.txt In their fifth season in the league, Inter Miami CF won
shapes: query=(1, 384) docs=(5, 384) latency=155.91 ms

Wrote 75 rows to reports/hw03/raw/ (retrieval_rows.jsonl, .csv) + technique_stats.json
Run finished: 2026-09-18T21:02:49.608037
```

The per query, per technique results are saved as machine readable files in reports/hw03/raw/

```
(data260) villyalsriprajeet@Mac data260-3319 % ls -la reports/hw03/raw/
total 0
drwxr-xr-x  5 villyalsriprajeet  staff   160 18 Sep 21:02 .
drwxr-xr-x  8 villyalsriprajeet  staff   256 18 Sep 21:02 ..
-rw-r--r--  1 villyalsriprajeet  staff  21739 18 Sep 21:02 retrieval_rows.csv
-rw-r--r--  1 villyalsriprajeet  staff  33897 18 Sep 21:02 retrieval_rows.jsonl
-rw-r--r--  1 villyalsriprajeet  staff   298 18 Sep 21:02 technique_stats.json
(data260) villyalsriprajeet@Mac data260-3319 %
```

Query to use

I wrote five questions in questions.yaml, each with an expected answer and expected source file. I committed questions.yaml to git before running any retrieval, so the answers couldn't be changed after seeing the output.

```
k: 5
questions:
- id: q1
  question: "Who were the champions of the 2023-24 Premier League?"
  expected_answer: "Manchester City"
  expected_source: "data/corpus/2023_24_premier_league.txt"
  single_source: true
- id: q2
  question: "Which club won the 2023-24 La Liga title?"
  expected_answer: "Real Madrid"
  expected_source: "data/corpus/2023_24_la_liga.txt"
  single_source: false
- id: q3
  question: "Which team won the 2023-24 EFL Championship and were promoted as champions?"
  expected_answer: "Leicester City"
  expected_source: "data/corpus/2023_24_efl_championship.txt"
  single_source: true
- id: q4
  question: "At which stadium was the 2024-25 UEFA Champions League final played?"
  expected_answer: "Allianz Arena"
  expected_source: "data/corpus/2024_25_uefa_champions_league.txt"
  single_source: true
- id: q5
  question: "Which club won the 2024 MLS Supporters' Shield?"
  expected_answer: "Inter Miami"
  expected_source: "data/corpus/2024_major_league_soccer_season.txt"
  single_source: false
```

Results

Technique	Chunks	Avg len (chars)	Top-1 cosine	Mean@5 cosine	Recall@5	Latency (ms)
token	499	700.3	0.6642	0.5945	1.00	39.99
semantic	167	1821.3	0.5838	0.5151	1.00	47.81
sentence_window	3202	95.0	0.6491	0.5153	1.00	147.70

```
(data260) villyalsriprajeet@Mac data260-3319 % python code/rag/make_tables.py
| Technique | Chunks | Avg chunk length (chars) | Top-1 cosine | Mean@5 cosine | Recall@5 | Mean retrieval latency (ms) |
|-----|-----|-----|-----|-----|-----|-----|
| token | 499 | 700.3 | 0.6642 | 0.5945 | 1.00 | 39.99 |
| semantic | 167 | 1821.3 | 0.5838 | 0.5151 | 1.00 | 47.81 |
| sentence_window | 3202 | 95.0 | 0.6491 | 0.5153 | 1.00 | 147.70 |

Wrote reports/hw03/METRICS.md
(data260) villyalsriprajeet@Mac data260-3319 %
```

Recall@5 is measured at the document level. All three reached 1.00, so the real difference is in the cosine scores and the latency.

On q3 (who won the 2023-24 EFL Championship), token chunking put a chunk at rank 1 with a high store score of 0.594, but that chunk was just a long list of past seasons with no answer in it, and its actual cosine was only 0.21. I think it scored high because it's full of the league name and lots of "YYYY-YY" season numbers, so it looks a lot like the question even though the answer isn't there.

Observations

Token chunking worked best for me. Its approx. 700 character chunks are big enough to hold an answer with some context but still stay focused, so it got the best cosine scores and was also the fastest. Semantic chunking made a few very large chunks, which seemed to bury the answer sentence inside a lot of other text and pulled the scores down. Sentence window split everything into single sentences, so a matching sentence can score well, but most chunks are tiny fragments like headings or citations, and searching that many vectors made it clearly the slowest.

Conclusion

For this corpus I'd go with token chunking. It had the best top 1 and mean cosine, full recall, and the lowest latency, and its medium sized chunks kept enough context without drowning the answer. Semantic chunks were too big and sentence window chunks were too small and too slow.

AI_USE

1. What I used an AI assistant for and what I did myself: I used an AI assistant to help set up the FastAPI auth app (Part 1) and the three technique LlamaIndex script (Part 2), and to work through some setup errors. I picked the domain and corpus sources, wrote the five questions, committed them before running, and ran everything and took the screenshots myself.

2. One AI output that was wrong/unsuitable: The first version of my corpus downloader only pulled about 35 KB because the API it used dropped all the tables, which was under the 200 KB minimum. My first package install also broke on my Intel Mac because the numpy and torch versions clashed.

3. How I detected the problem / verified the result: I caught the corpus problem from the script's own "35.2 KB - under 200 KB" message, and the install problem from the import error in the terminal. For the questions, I made sure each expected answer in its file before committing questions.yaml.

4. What I changed and why it works now: I changed the downloader to grab the rendered page and keep the tables as text, which got me to 303 KB. After that everything ran and the model embedded at dim 384.

Conclusion:

For HW3 I extended my existing data260-3319 repo with two parts. In Part 1 I built a FastAPI authentication app for my domain on port 8619, with the home, login, dashboard, and logout routes in a separate router, Starlette SessionMiddleware for a secure signed cookie, an idle session timeout, a protected dashboard, and Bootstrap styling, and I captured the set cookie header and the expired/logged out redirect as proof. In Part 2 I compared three LlamaIndex chunking techniques the token, semantic, and sentence window over a 303 KB corpus of public sports fixtures documents using the local all-MiniLM-L6-v2 model, committing my five questions before running, printing the embeddings and per technique retrieval tables, saving the raw results, and comparing them on cosine, recall, and latency, where token chunking came out best with the highest cosine scores, full recall, and lowest latency.